

Viva notes

Oil Price Shocks and India's Terms of Trade

Contents

00 · One-page summary	3
01 · Theory, applied to India	5
02 · Data: what we used, why, and what we fixed	9
03 · Methods: what each one does, why we chose it, and the traps	12
04 · Results in plain words	17
05 · Question bank with model answers	20
06 · Formula sheet	31

oo · One-page summary

Everyone should know this page cold. If you only have ten minutes before the viva, read this.

The 60-second version

India buys about 90% of the crude it refines from abroad, so every oil spike raises the price of what India imports relative to what it exports, the **net barter terms of trade** from lecture 21. We ask three things. Does an oil price rise worsen India's terms of trade, and by how much (H1)? Do rises and falls have the same effect (H2)? Does it matter *why* oil became expensive, supply or demand (H3)?

We use official data only (RBI, DGCI&S, PPAC, World Bank, UNCTAD, UN Comtrade, EIA, BIS) and the published shock series of Baumeister and Hamilton (2019) and Känzig (2021). Annual data run from FY1970-71 to FY2025-26, and monthly data from April 2019 to June 2026.

Answers.

- **H1, yes.** A 10% rise in the real oil price cuts the terms of trade by about 2.4% in the same year. The effect is fast (two to three months) but temporary: about 31% of the gap closes every year.
- **H2, no.** Rises and falls have about the same effect.
- **H3, no.** Demand-driven rises hurt at least as much as supply-driven ones, because a world boom also raises the prices of gold, coal, fertiliser and metals that India imports.
- **Exposure has roughly halved since the 1990s**, from -0.30 to -0.11 . This tracks the trade-share benchmark ($s_x - s_m$), because India became a big exporter of refined fuels (RCA of 4.37). Refining is India's hedge.

Numbers to know by heart

What	Number	Where it comes from
Short-run oil elasticity of ToT (annual, goods and services)	-0.24 (s.e. 0.049; robust 0.040)	t03, ARDL, N = 54
Speed of adjustment α	-0.31 (31% of the gap closes each year)	t03
Long-run elasticity	-0.09 , not significant ($p = 0.12$)	t03
Bounds test	$F = 4.74$; $p = 0.025$ vs the $I(0)$ bound, 0.070 vs the $I(1)$ bound	notebook 03
Short-run effect across six specifications	-0.20 to -0.28 , all significant	t04
Monthly elasticity two months after a rise	-0.39 (almost zero in month 0)	t08
Benchmark test coefficient (theory says 1)	1.70 , cannot reject 1 at 5% ($p = 0.07$)	t07
Rolling elasticity, window ending FY1989 → FY2024	$-0.30 \rightarrow -0.11$	t08b
Theory benchmark $s_x - s_m$, FY1999 → FY2025	$-0.25 \rightarrow -0.10$	RBI Table 111
NARDL short-run, rise vs fall	-0.27 vs -0.22 , symmetry $p = 0.62$	t06
H3, supply vs demand (annual G&S)	-0.07 vs -0.11 , equality $p = 0.71$	t09
RCA of refined petroleum, 1999 → 2025	0.06 → 4.37	UNCTADstat
Grubel-Lloyd for oil, one industry vs crude and products separately	0.55 vs 0.10	UN Comtrade
India's share of world oil demand (2024)	5.4%, third after the US and China	EIA
Merchandise ToT, Feb → Jun 2022 and Feb → Jun 2026	-16.1% and -13.1%	DGCI&S monthly

Fiscal years are labelled by the year they start: "FY2024" means 2024-25.

If they ask "what is new in your project?"

1. Indian studies look at the trade balance, inflation or the rupee. We model the **terms of trade itself**, which is the price ratio at the heart of the course's trade theory.

2. We test a **theory benchmark** built from the definition of the terms of trade, instead of only reporting a regression coefficient.
3. We separate **supply-driven and demand-driven** oil price changes with identified shocks that India cannot cause.
4. We show **exposure has halved** and explain it with RCA and Grubel–Lloyd, two measures from the lectures.
5. We **cleaned the official data**: four errors in RBI and DGCI&S tables were found and fixed.

Who takes which questions

Presenter	Leads on	Backs up
P1	Motivation, theory (offer curves, terms of trade, small vs large country), hypotheses	Policy
P2	Benchmark, RCA and Grubel–Lloyd, literature, data and the data problems	Theory
P3	Unit roots, ARDL, bounds test, diagnostics, local projections, benchmark test	Data
P4	NARDL (H2), supply vs demand (H3), rolling elasticity, policy, limitations	Methods

Anyone can take a question, but let the lead answer first and add one sentence at most. Two people giving long answers to the same question looks unprepared.

01 · Theory, applied to India

Each concept follows the same pattern: what the lecture says, how it applies to India's oil case, and the number from our data that goes with it. Lecture numbers (L05 to L22) refer to `course-material/lectures/`.

1. Terms of trade (L21)

What it is. With many goods, the terms of trade is the ratio of the export price index to the import price index. L21 calls it the commodity or net barter terms of trade:

$$NTT = (P_X / P_M) \times 100$$

In Indian data, the price indices are DGCI&S **unit value indices** (UVI): value divided by quantity, for a fixed basket of commodities. A rise in NTT is an improvement, because each unit of exports buys more imports.

Related measures, and why we don't use them.

- *Income terms of trade* = NTT × export volume index. It measures the import capacity of exports.
- *Gross barter terms of trade* = import volume / export volume.

Both need the DGCI&S **quantum** indices. The export quantum index on the 2012-13 base is erratic (monthly values from 54 to 2001), so we use price indices only.

Our two measures.

- **Merchandise NTT** (DGCI&S and RBI): goods only, the textbook definition.
- **Goods-and-services ToT** from the national accounts: the export deflator divided by the import deflator (World Bank WDI). It covers 1970 onwards and includes services such as software, which makes it a broader measure of India's purchasing power over imports.

2. Offer curves (L22): the core framework

What the lecture says.

- An offer curve shows how much of its export good a country is willing to give for different amounts of its import good. It is built from the trade triangles at each relative price, so it carries the country's production and preference information.
- Each offer curve bends towards the axis of the country's import good.
- Equilibrium is where the two curves cross. The terms of trade is the slope of the ray from the origin through that point.
- Shifts in the curves:
 - **Higher import demand by A:** trade volume rises and A's terms of trade fall.
 - **Higher import demand by B:** A's terms of trade improve.
 - **A tariff:** the tariff country's curve rotates towards its import axis. Trade volume falls and its terms of trade improve (if it is large).

Applied to India. Two "countries":

- India exports a composite good X (manufactures, services-linked goods, refined fuels) and imports oil Y.
- The rest of the world, including OPEC+, exports oil and imports X.
- With X on the horizontal axis and Y on the vertical axis, the ray's slope is Y/X: oil received per unit of exports, which is P_X/P_Y , India's terms of trade. A **flatter ray is worse** for India.

Case	What shifts	Prediction	When it happened
A. Supply cut (OPEC+ cut, war, sanctions)	The rest of the world offers less oil for each unit of X; its curve rotates towards the X axis. It is the L22 tariff case with the oil exporter doing the restricting.	India's ToT falls, a lot	1973, 1979, 1990, 2022, 2026

Case	What shifts	Prediction	When it happened
B. World demand boom	The rest of the world wants more oil <i>and</i> more of India's exports, so both effects work at once	Ambiguous in theory; this is our H3	2003–08
C. India's own demand grows	India offers more X for oil; India's curve shifts out	India's ToT falls (India is not a price-taker)	2000s and 2010s
D. Oil glut	The reverse of A	India's ToT improves	2014–16, 2020

Numbers.

- In our stylised diagram (slide 3), India's terms of trade fall from 1.00 to 0.86 in case A.
- The data: merchandise ToT fell 16.1% from February to June 2022 and 13.1% from February to June 2026.
- In 2014–16 (case D), the merchandise ToT rose 19.4%.

The elasticity point. India's short-run oil demand is very inelastic. When the rest of the world's curve shifts, almost all of the adjustment shows up in the price (ToT), not the volume.

- Our data: the elasticity of crude import volume to the oil price is only 0.16 (s.e. 0.09).
- The correlation between changes in the oil import bill and changes in the oil price is 0.96.

3. How big should the effect be? A benchmark from the definition

Start from $NTT = P_X / P_M$ and split both indices into oil and non-oil parts with value shares s_x (oil in exports) and s_m (oil in imports). Take logs and differentiate, holding non-oil prices fixed:

$$d \ln NTT \approx s_x \cdot d \ln P_{oil} - s_m \cdot d \ln P_{oil} = (s_x - s_m) \cdot d \ln P_{oil}$$

- **The sign:** India's oil share of imports (about 26% on average) is larger than of exports (about 10%), so the effect is negative. That is H1.
- **The size:** the benchmark gives the elasticity before any regression. It was -0.25 in FY1999 and is about -0.10 in FY2025.
- **Why it shrank:** s_x rose from about 3% in FY1990 to 12–22% since FY2008. The big private refineries at Jamnagar (1999, expanded in 2008) turned India into a large exporter of refined fuels.

What the regression adds. We test whether the actual effect matches the benchmark (coefficient 1.70; we cannot reject 1). Being a bit bigger makes sense, because oil also raises the prices of fertiliser, petrochemicals, plastics and freight, which the direct-share formula ignores.

4. Trade line, welfare and the small-country assumption (L11)

Welfare. With trade, a country consumes on its trade line, whose slope is the terms of trade. A worse ToT rotates the trade line inwards: the country reaches a lower community indifference curve even if production does not change. So a ToT loss is a **real-income loss**, not just a bookkeeping change.

- A rough size: net oil imports were 3.0% of GDP in FY2025 and 5.6% at the FY2012 peak. A 10% rise in oil prices costs roughly 0.3% of GDP in purchasing power at today's shares.

Small vs large country. A small country takes world prices as given. India is not small in oil: it uses 5.4% of world oil (EIA, 2024), third after the US and China.

- That is case C: India's own demand growth can push up the oil price and worsen its own terms of trade.
- For the regressions this means **endogeneity**: part of the oil price movement may be caused by India. That is why H3 uses shocks India cannot cause (Baumeister–Hamilton supply shocks; Känzig's OPEC news shocks).

5. Comparative advantage and RCA (Lo8)

Balassa's revealed comparative advantage:

$$RCA_{ij} = (X_{ij} / X_i) / (X_{wj} / X_w)$$

India's share of product j in its exports, divided by the world's share of j in world exports. RCA above 1 means India exports relatively more of j than the world does.

Applied.

- **Refined petroleum (SITC 334):** 0.06 in 1999, 1.26 in 2000, **4.37** in 2025.
- **Crude (SITC 333):** about 0.

India has almost no oil but a strong *acquired* comparative advantage in refining it. Lo5 even quotes India's advantage over China in "fuels". This is the main reason India's exposure to oil prices has fallen.

6. Economies of scale and monopoly power (L16)

Scale. Average cost $AC = F/Q + c$ falls with output when fixed costs are large. Refineries are a textbook case: Jamnagar is the world's largest refining complex. Scale is the source of the acquired advantage in \$5.

Monopoly markup. $MR = p(1 - 1/e) = MC$, so the markup $(p - MC)/p = 1/e$. The less elastic the demand, the bigger the markup.

- OPEC+ behaves like a dominant supplier. Short-run world oil demand is inelastic, so supply cuts raise the price sharply. This is case A in the offer-curve diagram.
- It also motivated H2: a cartel defends price floors by cutting output, so falls might be smaller than rises. The data do not show that.

Pro-competitive effect. More sellers mean a smaller markup. When India diversifies its crude suppliers (for example, buying more Russian crude after 2022), no single seller has as much market power over it.

7. Intra-industry trade: Grubel–Lloyd and the aggregation trap (L17, L19, L20)

Grubel–Lloyd index (L19):

$$GL = 1 - |X - M| / (X + M)$$

GL is 0 when trade is one-way (pure inter-industry) and 1 when exports equal imports (pure intra-industry).

The aggregation trap (L17): "the more broadly an industry is defined, the more trade appears to be intra-industry". India both imports and exports "mineral fuels" (HS chapter 27).

- Treat crude and products as one industry and GL is **0.55**: it looks like intra-industry trade.
- Measure crude (HS 2709) and products (HS 2710) separately and GL is **0.10**: it is really inter-industry.

Horizontal vs vertical (L20). Unit values separate the two. If export and import unit values are within $\pm 15\%$ of each other, the goods are horizontally differentiated (similar quality). Otherwise the trade is vertical.

- India's refined-export unit value is **1.23 to 1.41 times** its crude-import unit value per tonne (2005–2025), well outside $\pm 15\%$.
- So this is vertical, value-adding trade: India imports a raw input and exports a processed product. The refining margin is the hedge.

8. Partial and general equilibrium (L21)

Partial equilibrium looks at one market at a time. India's crude import demand is inelastic and the world supply curve shifts with OPEC+ decisions.

- A supply cut raises the price a lot and lowers the volume a little, so the **import bill jumps**.
- Our data: bill and price move together (correlation 0.96); volume barely responds (elasticity 0.16).

General equilibrium. Offer curves are general equilibrium: they bring in what happens in the other market too. In our regressions this is why we control for non-oil commodity prices: "changes in one market affect other markets".

Aggregation bias. L21's apples-and-oranges warning is why we split oil from non-oil trade and do not treat India's trade as one good.

9. Time horizons, distribution and endowments (L12, L13)

Time horizons (L13).

- **Very short run:** all factors fixed.
- **Medium run:** some factors specific (Ricardo–Viner).
- **Long run:** everything mobile (Heckscher–Ohlin–Samuelson).

Our error-correction model has exactly this structure. The short-run elasticity (-0.24) is the effect before anything adjusts; $\alpha = -0.31$ is how fast the economy adjusts; the long-run elasticity (-0.09) is what is left after adjustment.

Who gains and who loses (L13). In the Ricardo–Viner model, a higher oil price raises the return to factors specific to refining and hurts oil-using sectors (transport, fertiliser, airlines). Stolper–Samuelson: a change in relative prices helps the factor used intensively in the good whose price rose.

Endowments (L12). India is poorly endowed with oil, a natural-resource factor. So crude imports are structural, and policy can manage the exposure but cannot remove it.

10. Transport costs, CIF and FOB (L14)

- Imports are valued **CIF** (cost, insurance and freight).
- Exports are valued **FOB** (free on board).

So a jump in freight or war-risk insurance (Red Sea in 2024, Hormuz in 2026) raises measured import prices and lowers the measured ToT, even if the oil price itself did not change. We list this as a limitation.

11. Mercantilism and immiserizing growth (L06, L18)

Mercantilism (L06). A mercantilist would say the loss from an oil shock is the bigger trade deficit. Modern theory says the real loss is the fall in purchasing power, which is the ToT effect. The deficit is the financing side. Gold, a bullion import that India buys in large amounts, also moves India's import prices. That is why one robustness check controls for the gold price.

Immiserizing growth (L18, Bhagwati 1958). If India's growth raises its oil import demand enough to worsen its terms of trade (case C), part of the gain from growth leaks abroad. It needs very inelastic foreign supply, so it is a warning rather than India's actual situation.

12. Ideas outside our lecture set that examiners like to ask about

These are standard international economics. They are not in the 16 lectures we used, so keep the answers short.

- **Harberger–Laursen–Metzler effect.**
 - A worse ToT lowers real income; households smooth consumption, so saving falls and the current account worsens.
 - This links our ToT result to India's balance-of-payments stress in 1990–91, 2013 and 2022.
- **Prebisch–Singer hypothesis.**
 - Primary-commodity exporters face a long-run decline in their terms of trade against manufactures.
 - India is the opposite case: it *imports* the commodity. A long-run fall in commodity prices would help India, which is consistent with the merchandise ToT gains when oil collapsed in 2014–16.
- **Marshall–Lerner condition and J-curve.**
 - A depreciation improves the trade balance only if the export and import demand elasticities sum to more than one.
 - Oil demand is inelastic, which is why a weaker rupee does little for India's oil bill in the short run.
 - That is also why we model the terms of trade, not the trade balance.
- **Dutch disease.** A resource boom raises the real exchange rate and hurts other tradables. It applies to oil exporters. For India the mirror image is the rupee weakening in oil shocks (Ghosh 2011).
- **Real effective exchange rate (REER).** The trade-weighted relative price of Indian goods. We collected the BIS and RBI series but kept them out of the main models. The terms of trade is already a relative price, and the REER itself reacts to oil shocks, so controlling for it would absorb part of the effect we want to measure (a "bad control").

02 · Data: what we used, why, and what we fixed

Every raw file is listed with its web address and a checksum in `data/SOURCES.md`. The automatic checks are in `data/validation_report.md`. This page is the version to remember for the viva.

Our rule for sources

We used only **official statistical agencies** and the **published replication data of peer-reviewed papers**. No news sites, no blogs, no Kaggle or data aggregators. FRED was used only to cross-check the World Bank's Brent series.

The four datasets

Dataset	Period	Terms-of-trade measure	Used for
Monthly	Apr 2019 – Jun 2026 (87 months)	DGCI&S merchandise NTT, base 2012–13	Timing (local projections), 2022 and 2026 episodes, H3 monthly
Annual, long run	FY1970–71 – FY2025–26	Goods-and-services ToT, national accounts (WDI)	Main ARDL (H1), NARDL (H2), rolling elasticity, H3 annual
Annual, merchandise	FY1994–95 – FY2025–26	RBI/DGCI&S NTT, chain-linked across three bases	Robustness, benchmark test
Calendar-year theory data	1995/2000 – 2025	–	RCA, Grubel–Lloyd, unit values, India's share of world oil demand

Fiscal years. All Indian annual data are April to March and labelled by the starting year: FY2024 = 2024–25. Monthly world prices are averaged over April to March in our code, never mixed with calendar-year averages.

- We checked WDI's labelling: WDI's India "2024" equals RBI's FY2024–25 GDP, so WDI year t is FY $t/t+1$.
- The MUV index is calendar-year only, so the fiscal-year value is $0.75 \times \text{CY } t + 0.25 \times \text{CY } t+1$ (nine months of the fiscal year fall in t , three in $t+1$).

Variables and where they come from

Variable	Definition	Source
Merchandise NTT, monthly	Export UVI / import UVI $\times 100$, recomputed from the published indices	DGCI&S Terms of Trade files, base 2012–13
Merchandise NTT, annual	Same, fiscal-year indices, three bases chain-linked	RBI Handbook of Statistics 2025–26, Table 121
Goods-and-services ToT	Export deflator / import deflator (national accounts)	World Bank WDI
Oil price	Brent and Dubai; main series = average of Brent, Dubai and WTI	World Bank Pink Sheet (monthly since 1960)
Real oil price	Oil price / MUV index (price of manufactured exports), in 2010 prices	Pink Sheet and World Bank MUV
Indian crude basket	Weighted sour (Dubai/Oman) and sweet (Brent) grades India buys	PPAC (cross-check and the March 2026 hook)
Oil shares s_x, s_m	Oil exports / total exports; oil imports / total imports	RBI Table 111 (from FY1987)
Crude production and imports	Million tonnes	RBI Table 32
Rupee per dollar	Annual average and end of March	RBI Table 133
Non-energy commodity prices (control)	World Bank non-energy index / MUV	Pink Sheet
Gold price (robustness control)	US\$ per troy ounce, real	Pink Sheet

Variable	Definition	Source
REER	Real effective exchange rate	BIS broad index; RBI Table 135
Structural oil shocks	Supply, economic activity, consumption demand, inventory demand	Baumeister & Hamilton (2019), updated series to Mar 2026
Oil supply news shock	Surprise in oil futures around OPEC announcements	Känzig (2021, AER), updated to Dec 2025
RCA, refined and crude	Balassa index, SITC 334 and 333	UNCTADstat
Fuel trade by HS code	Values and tonnes, HS 2709 (crude) and 2710 (products)	UN Comtrade public API
Oil consumption by country	Thousand barrels a day	US EIA International Energy Data

About unit value indices. DGCI&S indices are Laspeyres indices with a fixed base (2012-13 = 100): the unit value (value / quantity) of each commodity, weighted by base-year trade shares. They are not pure prices, because a change in the quality or mix within a commodity code also changes the unit value. That is a known limitation, and it applies to every study of India's terms of trade: the official long-run series published by DGCI&S and the RBI are built from unit values.

Why these choices (short answers)

- **Why the national-accounts ToT for the long run?** It is the only Indian ToT series that goes back to 1970 on one consistent method, so it covers the 1973, 1979 and 1990 shocks. The merchandise series only links cleanly from FY1994.
- **Why deflate oil by MUV, not US CPI?** P_{oil} / MUV is the price of oil in terms of manufactures, which is the relative price on the axes of the offer-curve diagram. It is also the World Bank's standard deflator for commodity prices.
- **Why the average of Brent, Dubai and WTI, not the Indian basket?** The Indian basket is only published from 2020 in usable form, and PPAC's site does not even have FY2022-23. The average is available from 1960, and the results are the same with Brent alone.
- **Why not a quantum or volume index?** DGCI&S's 2012-13-base export quantum index is erratic (monthly values between 54 and 2001), which also makes RBI's published gross and income ToT for 2025-26 meaningless.
- **Why monthly data only from 2019?** DGCI&S's online archive of the 2012-13 base starts in FY2019-20, and the 1999-2000 base monthly page says "No Data Found". The long annual series carry the history.

How the merchandise series is chain-linked

RBI publishes NTT on three bases: 1978-79, 1999-2000 and 2012-13. They overlap but do not agree. We use the **newest base available for each year** and link the older bases at the overlap year with a ratio:

- 1999-2000 base for FY1999 to FY2011.
- 1978-79 base before FY1999, scaled so it equals the 1999 base in FY1999.
- 2012-13 base from FY2012 on, with the earlier segment scaled so the two meet in FY2012 (= 100).

This is the standard overlap-ratio method. It keeps the year-to-year growth rates of each base and only rescales the levels.

Problems we found in official data

This is a strong viva point: the data were checked, not just downloaded.

1. **RBI's base-year series contradict each other.** From FY1999-00 to 2007-08 the 1978-79-base NTT rises 21.7%, while the 1999-2000-base NTT falls 21.0%. The old base uses 1978-79 weights, from before India imported much oil or exported any refined fuel, so it misweights exactly the goods we study. We use the newest base for each period.
2. **Three published RBI NTT values disagree with RBI's own unit value indices** by more than rounding (FY2002-03, 2015-16, 2018-19). We recompute NTT from the indices, which is the textbook definition.
3. **The DGCI&S export quantum index is erratic** (see above). We never use quantum, gross or income ToT.
4. **WDI's 1960s deflators are a back-cast.** The export/import deflator ratio is frozen at one value through the 1960s, so the national-accounts ToT starts in FY1970.

5. **PPAC's website serves the wrong file for FY2022-23.** It is a provisional April 2023 table. We kept it, renamed it, and left that year of the Indian basket out.
6. **RBI Table 32 has a copy-paste error.** The rows for FY1990-91 to FY1997-98 are an exact copy of FY2000-01 to FY2007-08 in all four columns. We set those eight years to missing. Our validation script now looks for any repeated block of four or more rows in every table.
7. **No monthly DGCIS data before April 2019** are online. That limits the monthly sample to 87 months.

Checks that pass

- NTT recomputed from the unit value indices equals the published DGCIS NTT in every month, on both bases.
- DGCIS fiscal-year indices equal RBI Table 121 for FY2019-20 to FY2025-26.
- World Bank Brent vs FRED/EIA Brent: correlation 0.9998, mean difference \$0.33 a barrel over 441 months.
- The PPAC Indian basket lies between Dubai and Brent in 95% of normal months (it diverged in March 2026, when Gulf sour grades spiked).
- WDI and UNCTAD terms of trade are identical where both exist.
- RBI oil imports: Table 111 equals Table 115.
- Crude import tonnes: UN Comtrade and RBI Table 32 agree within $\pm 9\%$.

Quick facts from the data

	Value
Oil share of India's imports, FY1987–2025	mean 26%, range 15–37%
Oil share of India's exports, FY1987–2025	mean 10%, range 0.1–22% (22% in FY2022)
Crude imports as share of crude refined, FY2025-26	90% (246 of 272 million tonnes)
Net oil imports, % of GDP	5.6% at the FY2012 peak, 3.0% in FY2025
Monthly merchandise NTT, Apr 2019–Jun 2026	mean 111.9; high 163.2 (May 2020, oil crash); low 89.4 (Jun 2022)
Brent, Apr 2019–Jun 2026	mean \$74; low \$23 (Apr 2020); high \$120 (Apr 2026)
March 2026	Brent \$104, Indian basket \$113, rupee 94.65 per dollar at end-March

03 · Methods: what each one does, why we chose it, and the traps

For each method you need to know four things: what it does, why we used it rather than the alternative, what it assumes, and how to read our output. The questions examiners like most are marked **Trap**.

Notation: $y = \ln \text{ToT}$, $x = \ln \text{real oil price}$, $z = \ln \text{real non-energy commodity price}$. Δ is the first difference.

Logs and elasticities

Every variable is in logs, so each coefficient is an **elasticity**: the percentage change in ToT for a 1% change in the oil price. "−0.24" means a 10% oil rise lowers the ToT by 2.4%.

- In the monthly models we use $100 \times \ln$, so the coefficients read directly as percentages.

Step 1. Unit-root tests (notebook 03, backup slide)

Why. If two series both trend, regressing one on the other can give a high R^2 and "significant" coefficients even when they are unrelated (a **spurious regression**). We need to know each series' order of integration.

Test	Null hypothesis	Reject the null means
ADF (augmented Dickey–Fuller)	Unit root	Stationary
Phillips–Perron	Unit root (robust to serial correlation and heteroskedasticity)	Stationary
KPSS	Stationary	Unit root
Zivot–Andrews	Unit root, allowing one structural break	Stationary with a break

Our results (p-values).

Series	ADF level	ADF diff	KPSS level
$\ln \text{ToT goods and services}$	0.057	0.000	0.10
$\ln \text{ToT merchandise}$	0.27	0.001	0.01
$\ln \text{real oil price}$	0.053	0.000	0.013
$\ln \text{real non-energy prices}$	0.36	0.000	0.10

- Reading:** levels are borderline between $I(0)$ and $I(1)$; every first difference is clearly stationary. **Nothing is $I(2)$.**
- Why that suits ARDL:** the bounds test is valid whether each regressor is $I(0)$ or $I(1)$, so we do not have to take a stand on the borderline cases.
- Zivot–Andrews** does not reject a unit root in levels even with a break (break years: ToT 1985, oil 1999), so a single break is not driving the results.

Trap. "Why run both ADF and KPSS?" Because they have opposite nulls. ADF has low power in short samples, so failing to reject a unit root is weak evidence. KPSS is the confirmation from the other side.

Step 2. ARDL and the bounds test, for H_1 (notebook 03)

The model. Lag lengths were chosen by AIC with a maximum of 2, which gave ARDL(1,1,1). In error-correction form (UECM):

$$\Delta y_t = c + \alpha \cdot y_{t-1} + \beta_1 \cdot x_{t-1} + \beta_2 \cdot z_{t-1} + \gamma_1 \cdot \Delta x_t + \gamma_2 \cdot \Delta z_t + \varepsilon_t$$

- Short-run elasticity:** γ_1 , the effect of this year's oil change on this year's ToT change.
- Long-run elasticity:** $-\beta_1/\alpha$. Its standard error comes from the delta method, because it is a ratio of two estimates.
- Speed of adjustment:** α . It must be negative and between -1 and 0 . $\alpha = -0.31$ means 31% of any gap from the long-run relationship closes in a year (a half-life of about two years).

The bounds test (Pesaran, Shin and Smith 2001). An F-test of $\alpha = \beta_1 = \beta_2 = 0$, "no long-run relationship". The critical values come in two bounds:

- the **lower bound** assumes all regressors are $I(0)$;
- the **upper bound** assumes all are $I(1)$.

F compared with	Conclusion
Above the upper bound	A long-run relationship exists
Below the lower bound	No long-run relationship
Between the bounds	Inconclusive

We use **case 3** (unrestricted intercept, no trend) and **finite-sample** critical values (Kripfganz and Schneider 2020), because 54 observations is small and the asymptotic tables are too lenient.

Our result. $F = 4.74$.

Significance	Lower bound	Upper bound
10%	3.32	4.32
5%	4.02	5.13
1%	5.69	7.03

- F is above the 10% upper bound, so there is a long-run relationship at 10% ($p = 0.070$ against the $I(1)$ bound).
- F lies between the bounds at 5%, so the 5% test is inconclusive ($p = 0.025$ against the $I(0)$ bound).
- The error-correction coefficient is strongly significant ($t = -3.60$), which also points to adjustment towards a long-run level.

Diagnostics.

Test	p-value	Reading
Breusch–Godfrey (2 lags)	0.90	No serial correlation
Breusch–Pagan	0.019	Heteroskedasticity , so we also report HC1 robust s.e.
Jarque–Bera	0.26	Residuals are normal
Ramsey RESET	0.84	Functional form is fine
CUSUM and CUSUMSQ	inside 5% bands	Coefficients are stable over time

- With HC1 robust standard errors the short-run s.e. is 0.040 instead of 0.049. The conclusion does not change.
- HC1 is White's robust covariance with a small-sample correction, $n/(n - k)$.

Robustness (six specifications). The baseline, oil only, adding gold, Brent instead of the average crude, post-1980 only, and merchandise ToT. The short-run elasticity stays between -0.20 and -0.28 and is significant in every one.

Why ARDL and not Johansen or Engle–Granger?

- It works with a mix of $I(0)$ and $I(1)$ variables. Johansen needs everything to be $I(1)$.
- It behaves better in small samples ($N = 54$).
- It is a single equation with an obvious dependent variable (ToT). We are not interested in modelling the oil price as a function of India's ToT.
- It gives short-run and long-run effects in one regression.

Traps.

- "Your bounds test isn't significant at 5%, so where is your long-run relationship?" Agree: the long run is weak, the long-run elasticity (-0.09) is not significant, and our headline is the **short-run** effect. It is significant in every specification and does not depend on cointegration.
- "Is the oil price exogenous?" Mostly yes for annual data, because India's own demand moves world prices only slowly. We address it directly in H_3 with shocks India cannot cause.
- "Why no trend?" The ToT has no significant linear trend over 1970–2025: it is 110 in FY1970 and 98 in FY2025, and a trend term has $p = 0.12$. Adding one would use up a degree of freedom we cannot spare.

Step 3. Nonlinear ARDL, for H2 (notebook 04)

Idea (Shin, Yu and Greenwood-Nimmo 2014). Split the oil price into two running totals, one of all the increases and one of all the decreases:

$$x^+_t = \sum \max(\Delta x_j, 0) \text{ and } x^-_t = \sum \min(\Delta x_j, 0)$$

Then estimate an ARDL with both:

$$\Delta y_t = c + \alpha \cdot y_{t-1} + \theta^+ \cdot x^+_{t-1} + \theta^- \cdot x^-_{t-1} + \pi^+ \cdot \Delta x^+_t + \pi^- \cdot \Delta x^-_t + \text{controls} + \varepsilon_t$$

- Long-run effects are $L^+ = -\theta^+/\alpha$ and $L^- = -\theta^-/\alpha$.
- **Wald tests:** long-run symmetry $L^+ = L^-$, short-run symmetry $\pi^+ = \pi^-$.

Reading the signs. x^- only ever goes down. So a **negative** coefficient on x^- means an oil price *fall* **raises** the ToT. Both coefficients negative means "rises hurt and falls help". Asymmetry is about whether the two have the same size.

Our result (baseline, N = 54).

- Short run: rise -0.27 , fall -0.22 , symmetry $p = 0.62$.
- Long run: rise -0.16 , fall -0.23 , symmetry $p = 0.035$.

Is the long-run asymmetry real? No, it is fragile:

- $p = 0.05$ with an extra lag;
- $p = 0.08$ from FY1975;
- $p = 0.28$ from FY1980;
- reversed and not significant for merchandise ToT.

Short-run symmetry is never rejected in any specification. So we conclude the effects are **symmetric**, and H2 is not supported. Notice that the baseline's long-run asymmetry goes the *opposite* way to Mork (1989): falls help more than rises hurt. It comes from the 1970s.

Step 4a. Benchmark test (notebook 05)

$$\Delta \ln \text{NTT}_t = a + b \cdot [(s_x - s_m)_{t-1} \cdot \Delta \ln P_{\text{oil},t}] + c \cdot \Delta \ln P_{\text{nonoil},t} + e_t$$

- Theory says **b = 1**: the ToT moves exactly by the oil share gap times the oil price change.
- Shares are lagged one year, so they are not affected by this year's oil price.
- Data: merchandise NTT, FY1995–FY2024, N = 30, HAC (Newey–West) standard errors.

Result. $b = 1.70$ (s.e. 0.38).

- It is significantly different from 0 ($p < 0.001$).
- It is not significantly different from 1 at 5% ($p = 0.07$).
- The theory gets the sign and the order of magnitude right. The extra effect is the indirect channels: fertiliser, petrochemicals, plastics and freight.

Step 4b. Rolling 20-year elasticity (notebook 05)

$\Delta \ln \text{ToT}$ on $\Delta \ln P_{\text{oil}}$ and $\Delta \ln P_{\text{non-energy}}$, re-estimated on every 20-year window, with HC1 standard errors.

- **Result:** -0.30 for the window ending FY1989; about -0.25 to -0.27 through the 2000s; -0.16 by FY2015; -0.11 for the window ending FY2024.
- From about 2013 it runs close to the theory benchmark.
- **Why 20 years?** Long enough to include several oil cycles in each window, short enough to show change.

Step 4c. Local projections, monthly timing (notebook 05)

Jordà (2005). Instead of one model that is iterated forward (like a VAR), run one regression per horizon h :

$$\ln \text{NTT}_{t+h} - \ln \text{NTT}_{t-1} = a_h + \beta_h \cdot \Delta \ln \text{Brent}_t + \varphi_h \cdot \Delta \ln \text{Brent}_{t-1} + \psi_h \cdot \Delta \ln \text{NTT}_{t-1} + u_{t+h}$$

- β_h is the cumulative response of the ToT h months after a 1% oil price change.
- $h = 0, 1, \dots, 6$. HAC standard errors with $h + 1$ lags, because overlapping horizons create serial correlation.
- 79 to 85 monthly observations (April 2019 to June 2026).

Result.

h (months)	0	1	2	3	4	5	6
β_h	-0.02	-0.23	-0.39	-0.37	-0.28	-0.30	-0.37

Nothing happens in month 0. The effect builds in one to three months and stays.

Why local projections and not a VAR? With only about 85 months, a VAR's impulse responses depend heavily on getting the whole lag structure right. Local projections estimate each horizon directly, so a mistake at one horizon does not carry over to the others. The cost is wider confidence bands, which we accept.

Step 5. Supply-driven vs demand-driven oil prices, for H3 (notebook 06)

Step 1, split each oil price change. Regress the monthly oil price change (February 1975 to March 2026, 612 months) on Baumeister and Hamilton's four structural shocks, each with two lags:

- oil supply;
- economic activity;
- oil consumption demand;
- oil inventory demand.

$R^2 = 0.80$, so the four shocks explain most oil price movements.

- **Supply part:** the fitted contribution of the supply shocks.
- **Demand part:** the fitted contribution of the other three shocks.

Step 2, compare. Regress ToT changes on the two parts and test whether their coefficients are equal. This is done in three samples:

- annual goods-and-services ToT (fiscal-year sums of the monthly parts);
- annual merchandise ToT;
- monthly merchandise ToT, as three-month cumulative effects.

Result. Demand-driven rises hurt at least as much in all three. Equality is never rejected ($p = 0.54$ to 0.71).

Sample	Supply-driven	Demand-driven	p (equal)
Annual G&S, FY1975–2025	-0.07 (0.07)	-0.11 (0.05)	0.71
Annual merchandise, FY1995–2024	-0.03 (0.10)	-0.13 (0.09)	0.55
Monthly, 3-month cumulative, 2019–26	-0.24 (0.17)	-0.37 (0.09)	0.54

Cross-check with Känzig (2021). These are oil supply *news* shocks, measured from oil futures prices in a narrow window around OPEC announcements. India cannot cause them, so they settle the endogeneity worry. One shock raises oil prices about 10% on impact.

- **Annual:** ToT falls 0.95% per shock (s.e. 0.49), an elasticity of about -0.1.
- **Monthly:** ToT falls 4.2% at two months and 4.7% at three months, an elasticity of about -0.4 to -0.5.

Pure supply shocks clearly hurt India, but not more than demand shocks.

Why Baumeister–Hamilton and not Kilian (2009)? Kilian assumes the oil supply curve is completely vertical within a month. Baumeister and Hamilton replace that strict assumption with Bayesian priors on the elasticities. They find supply shocks matter more than Kilian's method suggests, which made them the tougher test for H3.

Traps.

- "Your supply and demand parts are estimated, so aren't your second-step standard errors too small?" Yes, in principle. That is the generated-regressor problem. But the first stage is very precise (612 months, $R^2 = 0.80$), and our conclusion is that the two effects are *not* different. Larger standard errors would only make that conclusion stronger.
- "The BH shocks are about the US/world economy. Why do they apply to India?" They identify *why* the *world* oil price moved. India takes that price from the world market, so the source of the world price move is exactly what we need.

What we deliberately did not do

- **No VAR for the annual data.** With 54 years, a VAR with three or four variables and two lags uses up too many degrees of freedom. ARDL is more economical.
- **No GARCH or volatility models.** Our question is about the level of the terms of trade, not its volatility.
- **No quantum-index measures** (income or gross ToT), because the DGCI&S export quantum index is unreliable.
- **No trade-balance regression.** The trade balance mixes prices and volumes and depends on income and the rupee. The terms of trade is the pure price channel that trade theory is about.

04 · Results in plain words

How to say each result, what it means economically, and where the number comes from. Table names (to3 and so on) refer to output/tables/.

H1: oil price rises worsen India's terms of trade. Supported.

Say it like this: "A 10% rise in the real oil price lowers India's terms of trade by about 2.4% in the same year. The effect is fast but temporary: about a third of it unwinds every year."

Estimate	Value	s.e.	p
Short-run elasticity	−0.241	0.049 (HC1 0.040)	< 0.001
Long-run elasticity	−0.092	0.059	0.12
Long-run non-energy commodity elasticity	−0.249	0.177	0.16
Error-correction α	−0.308	0.086	0.001

- **Statistical significance:** the short-run effect is significant at well below 1%, with or without robust standard errors, and in all six specifications (−0.20 to −0.28).
- **Economic significance:**
 - The 2022 oil rise (+83% in the fiscal-year average from FY2020 to FY2022) maps into roughly $0.24 \times 83 \approx 20\%$ by the elasticity alone. The actual fall was 14% (goods and services) and 24% (merchandise).
 - So the elasticity gives the right order of magnitude for a real episode.
- **Why the long run is small:** the economy adjusts. Export prices of refined fuels follow crude with a lag, the rupee and export mix adjust, and oil spikes tend to reverse. That matches the time-horizon idea from L13: the short-run effect is bigger than the long-run effect.

Monthly timing (to8). "Almost nothing happens in the month oil rises. Two months later the terms of trade are about 0.4% lower for every 1% rise in Brent, and they stay there."

- **Why the lag:** crude cargoes are priced on the previous month's benchmarks and recorded at customs when they arrive, often weeks after loading.

The 2026 episode.

- From February to April 2026, export unit values rose 18.1% and import unit values 15.7%. Refined-fuel export prices rose with crude, so the ToT barely moved at first.
- By June the ToT was 13.1% below February, as import prices kept rising (+20% Feb–Jun) while export prices fell back.
- This is the refining hedge working, but only for a while.

The size: theory benchmark and exposure over time

Benchmark test (to7).

- "Theory predicts a coefficient of 1. We get 1.70 and cannot reject 1 at 5%. Simple trade-share arithmetic from the definition of the terms of trade explains the direction and the size."
- The plain merchandise elasticity over the same years is −0.30, against an average benchmark of −0.16. The difference is the indirect channels: fertiliser, petrochemicals, plastics and freight.

Rolling elasticity (to8b).

- "India's exposure has roughly halved. A 10% oil rise used to cost about 3% of the terms of trade, and now it costs about 1.1%."
- **Why:** India's oil share of exports rose from about 3% (FY1990) to 12–22% (since FY2008). The benchmark fell from −0.25 to −0.10 over FY1999–2025, and the estimated elasticity followed it.
- **Supporting theory numbers:**
 - RCA in refined petroleum: $0.06 \rightarrow 4.37$.
 - Grubel–Lloyd for oil: 0.55 aggregated vs 0.10 separated.
 - Export/import unit value ratio: 1.23 to 1.41.

H2: rises and falls have unequal effects. Not supported.

Say it like this: "In the short run, a rise and a fall of the same size move the terms of trade by about the same amount. The only asymmetry we find is in the long run, it depends on the 1970s, and it goes the opposite way to what we expected."

	Rise	Fall	p (symmetry)
Short run, baseline	−0.27	−0.22	0.62
Short run, all 7 specifications			0.54 – 0.91
Long run, baseline	−0.16	−0.23	0.035
Long run, from FY1980	−0.11	−0.15	0.28

- **What it means:** India gains from cheap oil about as much as it loses from expensive oil. Policy should be symmetric too: save in the slumps to spend in the spikes.
- **Why the expectation failed:** Mork's asymmetry is about *output*. Output can respond asymmetrically because firms delay investment when prices are uncertain. The terms of trade is a *price ratio* that passes oil prices through almost mechanically in both directions.

H3: supply-driven rises hurt more than demand-driven ones. Not supported.

Say it like this: "It matters *how much* oil rises more than *why*. Demand-driven rises hurt at least as much, because a world boom raises the price of everything India imports, not just oil."

Sample	Supply-driven	Demand-driven	p (equal)
Annual G&S, FY1975–2025	−0.07	−0.11	0.71
Annual merchandise, FY1995–2024	−0.03	−0.13	0.55
Monthly, 3-month cumulative	−0.24	−0.37	0.54

- **Why the offer-curve intuition (case B) fails for India:** in a boom, India's other imports also get dearer (gold, coal, fertiliser, edible oils, metals). India's exports are mostly manufactures and services, whose prices rise less. So the rest of the world's offer curve moves against India on several goods at once.
- **The Känzig check:** pure OPEC news shocks lower the ToT by about 1% per shock annually and 4–5% after two to three months. Supply shocks clearly hurt, just not more than demand shocks.
- **Policy meaning:** it does not help India to ask whether a spike is "OPEC's fault" or a "global boom". The buffers are needed either way.

Big-picture episodes (t10, fiscal-year averages)

Episode	Real oil	ToT goods and services	ToT merchandise
1973 OPEC embargo (FY1972 → FY1974)	+284%	−38.6%	–
1979 Iranian revolution (FY1978 → FY1980)	+110%	−19.2%	–
1990 Gulf war (FY1989 → FY1990)	+21%	−9.8%	–
2008 price spike (FY2006 → FY2008)	+20%	+4.9%	+5.8%
2014–16 collapse (FY2013 → FY2015)	−50%	−1.0%	+19.4%
2020 COVID crash (FY2019 → FY2020)	−26%	+4.7%	+9.6%
2022 Russia–Ukraine (FY2020 → FY2022)	+83%	−14.4%	−23.7%

2008 looks odd. The spike lasted only months (Brent peaked in July 2008 and crashed by December), so the fiscal-year average barely captured it. Monthly data would show it, but DGCIS monthly data for 2008 are not online. This is one reason we also use monthly data where we can.

Where every number on the slides comes from

Slide	Number	Source
1 Hook	Brent +46% (\$71.1 → \$103.7), basket \$69 → \$113, Rs 94.65, ToT −13.1%	Pink Sheet; PPAC; RBI Table 133; DGCI&S (notebook 01)
2 Hypotheses	90% of crude imported; 5.4% of world demand	RBI Table 32; EIA (notebook 02)
3 Offer curves	1.00 → 0.86 (illustrative); −16% (2022), −13% (2026)	Diagram code; DGCI&S monthly
4 Benchmark	−0.25 → −0.10; RCA 0.06 → 4.37; GL 0.55 vs 0.10	RBI Table 111; UNCTADstat; UN Comtrade (notebook 02)
6 Data	+21.7% vs −21.0%; Table 32 duplicate; QI 54 to 2001	Validation report
8 H1	−0.24, α −0.31, LR −0.09, F 4.74 (p 0.07), range −0.20 to −0.28	to3, to4
9 Timing	−0.39 at 2 months; benchmark 1.70 (p 0.07); +18% vs +16%	to8, to7, notebook 01
10 H2	−0.27 vs −0.22; p 0.54–0.91; p 0.28 from 1980; −0.30 → −0.11	to6, to8b
11 H3	p 0.54–0.71; Känzig 4–5%	to9, notebook 06
12 Conclusion	2.4%, halved	to3, to8b

05 · Question bank with model answers

140 questions we could be asked, with answers of a length you can actually say out loud. The tag after each question is who answers first (P1 to P4; "Any" means whoever is closest).

Don't memorise the wording. Learn the point and the number, then say it your own way. Examiners can tell when an answer is recited.

Sections

- A. The big picture
 - B. Theory
 - C. Literature
 - D. Data
 - E. Econometrics
 - F. Results
 - G. Policy and current affairs
 - H. Limitations and extensions
 - I. Curveballs
 - J. When you don't know the answer
-

A. The big picture

A1. Why did you choose this topic? (P1)

India imports about 90% of the crude it refines, and oil is the largest single item in its import bill, about a quarter of all imports on average. Almost every oil spike has come with stress on India's external accounts: 1990 before the 1991 crisis, 2013, 2022, and now March 2026. So the link between oil and what India's exports can buy is one of the most practical questions in Indian international economics.

A2. What is the terms of trade, in one line? (P1)

The price of what a country sells relative to the price of what it buys. We use the net barter terms of trade from lecture 21: the export unit value index divided by the import unit value index, times 100.

A3. Why study the terms of trade and not the trade balance? (P1)

The trade balance mixes prices, volumes, income and the exchange rate, so it is hard to say what an oil shock does to it on its own. The terms of trade is the pure price channel, and it is what the trade theory in our course is about: the offer-curve equilibrium is a terms-of-trade equilibrium. It is also the welfare channel: a worse terms of trade is a real-income loss.

A4. What is your main finding in one sentence? (P1)

A 10% oil price rise cuts India's terms of trade by about 2.4% within a year, mostly within two to three months, whatever the cause of the rise, and India is about half as exposed as it was in the 1990s because it became a big refined-fuel exporter.

A5. What is new in your work? (P2)

Indian studies model the trade balance, inflation or the rupee; we model the terms of trade directly. We test it against a benchmark derived from the definition of the terms of trade, we separate supply-driven from demand-driven oil prices with identified shocks, we explain the fall in exposure with RCA and Grubel-Lloyd, and we cover 2022 and 2026.

A6. Why India? Wouldn't any oil importer look the same? (P1)

India is unusual in two ways. It is a large buyer (third-largest consumer, 5.4% of world demand), so it is not a pure price-taker. And it is a big exporter of refined fuels (RCA of 4.37), so part of every oil shock comes back to it through export prices. That hedge is what makes the size of India's exposure an empirical question.

A7. Is a fall in the terms of trade always bad? (P1)

Not always. If export prices fall because India became more productive, it can still gain overall. But a fall caused by an oil price spike is a pure loss: nothing improved in India, it simply has to give more exports for the same oil.

A8. What period do you cover, and why? (P2)

Annual data from FY1970-71 to FY2025-26, the longest consistent Indian series, which includes the 1973, 1979, 1990, 2008, 2014-16, 2020 and 2022 episodes. Monthly data from April 2019 to June 2026, which is all that DGCI&S publishes online, and covers COVID, 2022 and the 2026 Hormuz shock.

A9. What exactly happened in March 2026? (P1)

After the US and Israel struck Iran at the end of February, traffic through the Strait of Hormuz fell from about 20 million barrels a day to a trickle. The IEA's March Oil Market Report called it the largest oil supply disruption in

history. Brent rose 46% in a month, from \$71 to \$104, India's crude basket went from \$69 to \$113, and the rupee ended March at 94.65 per dollar.

A10. Who would use your results? (P4)

Anyone who has to plan for oil shocks: the Finance Ministry setting fuel taxes, the RBI judging how much a spike will hurt the external accounts, and oil companies planning stocks. The rule of thumb is simple: roughly 0.25% of the terms of trade lost per 1% oil rise in the short run, with the loss mostly in place within three months.

B. Theory

B1. Explain an offer curve. (P1)

It shows how much of its export good a country is willing to give in exchange for different amounts of its import good. It comes from the trade triangles at each relative price, so it already contains the country's production possibilities and preferences. Where India's and the rest of the world's curves cross is the trade equilibrium, and the slope of the ray from the origin to that point is the terms of trade.

B2. Why does India's offer curve bend towards the oil axis? (P1)

Because each extra unit of exports costs India more (increasing opportunity cost) and each extra unit of oil is worth less to it (diminishing marginal utility). So India only offers more exports if it gets increasingly more oil per unit, and the curve becomes steeper. Every offer curve bends towards the axis of the country's import good.

B3. How is an OPEC+ supply cut like a tariff? (P1)

In lecture 22, a tariff makes the imposing country offer less trade at every price; its offer curve rotates towards its import axis, the volume of trade falls and its terms of trade improve if it is large. A supply cut is the oil exporter doing exactly that: it offers less oil for every unit of India's exports. So the oil exporter's terms of trade improve and India's fall.

B4. Where is the terms of trade in your diagram? (P1)

With India's exports on the horizontal axis and oil on the vertical axis, the ray's slope is oil per unit of exports, which is P_X/P_Y . A flatter ray is worse for India. In our stylised diagram, the supply cut flattens it from 1.00 to 0.86.

B5. What happens in a world demand boom? (P1)

Two things at once. The rest of the world wants more oil, which pushes oil prices up, and it also wants more of India's exports, which pushes India's export prices up. In the diagram both curves move, so theory cannot sign the net effect. That is why H3 is an empirical question.

B6. Is India a small country or a large country? (P1)

Large, in oil. It uses 5.4% of the world's oil, third after the US and China, so growth in India's demand can raise the world price. That is case C on our slide. It means part of the oil price may be caused by India, which is why we use shocks India cannot cause for H3.

B7. What is the welfare cost of a worse terms of trade? (P1)

In lecture 11, a country consumes on its trade line, whose slope is the terms of trade. A worse terms of trade rotates the line inwards, so the country reaches a lower indifference curve even if it produces the same. As a rough size, net oil imports are about 3% of GDP, so a 10% oil rise costs India around 0.3% of GDP in purchasing power.

B8. Derive your benchmark. (P2)

Write the export and import price indices as weighted averages of oil and non-oil prices, with oil weights s_x and s_m . Take logs and differentiate, holding non-oil prices fixed: $d \ln P_X \approx s_x d \ln P_{oil}$ and $d \ln P_M \approx s_m d \ln P_{oil}$. Subtract: $d \ln NTT \approx (s_x - s_m) d \ln P_{oil}$. With $s_m \approx 0.26$ and $s_x \approx 0.10$ on average, it is negative.

B9. Your benchmark coefficient is 1.70, not 1. Is the theory wrong? (P2)

No. We cannot reject 1 at 5% ($p = 0.07$). It is a bit above 1 because the formula only counts oil itself, while oil also raises the price of fertiliser, petrochemicals, plastics and freight, which India imports. Those indirect channels make the real effect somewhat larger than the direct share.

B10. What is RCA, and what did you find? (P2)

Balassa's index: a product's share in India's exports divided by its share in world exports. Above 1 means India has a revealed comparative advantage. For refined petroleum it went from 0.06 in 1999 to 1.26 in 2000 and 4.37 in 2025, so refined fuels are over four times as important in India's exports as in world exports. For crude it is about zero.

B11. How can India have a comparative advantage in refining with no oil of its own? (P2)

It is an acquired advantage, not one from natural endowments. Refining has huge fixed costs, so scale cuts average cost (lecture 16), and India has some of the largest and most complex refineries in the world on its west coast, close to Gulf crude and to export markets. The scarce factor is capital and know-how, not crude.

B12. What does the Grubel-Lloyd result show? (P2)

When crude and refined products are lumped together as "mineral fuels", GL is 0.55, which looks like a lot of intra-industry trade. When we measure crude and products separately it falls to 0.10. That is exactly lecture 17's warning: the broader the industry definition, the more trade looks intra-industry. India's oil trade is really inter-industry: crude in, fuels out.

B13. Is India's oil trade horizontal or vertical? (P2)

Vertical. Lecture 20's rule says trade is horizontal if export and import unit values are within $\pm 15\%$ of each other. India's refined exports are worth 1.23 to 1.41 times its crude imports per tonne, well outside that band. India imports a raw input and sells a processed, higher-value product.

B14. What does the partial equilibrium model say about the oil import bill? (P2)

India's demand for crude is very inelastic in the short run. When the world supply curve shifts up, the price rises a lot and the volume barely falls, so the import bill jumps. In our data the bill moves almost one-for-one with the price (correlation 0.96), and the volume elasticity is only 0.16.

B15. How does the monopoly model from lecture 16 apply? (P1)

A monopolist sets marginal revenue $p(1 - 1/\epsilon)$ equal to marginal cost, so the markup is 1 over the elasticity of demand. Short-run oil demand is very inelastic, so a dominant supplier like OPEC+ can raise prices a lot by restricting output. That is the supply-shock case in our offer-curve diagram.

B16. If oil demand is so inelastic, why doesn't OPEC+ raise prices forever? (P1)

Because the demand OPEC+ faces is the residual demand after other producers, and that is much more elastic. High prices bring in US shale and other non-OPEC supply. Demand also becomes more elastic over time as people switch fuels and cars, and cartel members cheat on quotas. A monopolist always ends up on the elastic part of its own demand curve.

B17. How does diversifying suppliers help India? (P4)

That is the pro-competitive effect in lecture 16: more sellers means less market power and a smaller markup. When India buys from more countries, as it did by shifting a large share of its purchases to Russian crude after 2022, no single seller can squeeze it as much, and it can buy at a discount.

B18. How does Heckscher–Ohlin fit in? (P2)

In H-O, countries import goods that use their scarce factors intensively. India is poorly endowed with oil, a natural resource, so crude imports are structural. That is why the exposure can be managed, through refining, reserves and diversification, but not removed.

B19. How do the time horizons in lecture 13 relate to your model? (P3)

Lecture 13 separates the very short run, where everything is fixed, from the medium run (specific factors) and the long run (everything mobile). Our error-correction model has the same structure: the short-run elasticity of -0.24 is the effect before anything adjusts, $\alpha = -0.31$ is how fast adjustment happens, and the long-run elasticity of -0.09 is what is left afterwards.

B20. Who gains and who loses inside India when oil rises? (P4)

In the specific-factors model, the factors tied to refining gain, because refined-product prices rise with crude. Sectors that use oil heavily lose: transport, airlines, fertiliser, petrochemical users. Consumers lose purchasing power, and the government faces a choice between higher prices, lower fuel taxes or bigger subsidies.

B21. Isn't the real problem of an oil shock the trade deficit? (P1)

That is the mercantilist view from lecture 6. The deficit is how the loss is financed. The real loss is that each unit of India's exports buys less oil, which is the terms-of-trade effect. A deficit can be financed by borrowing; the lost purchasing power cannot be borrowed back.

B22. What is immiserizing growth, and does it apply here? (P1)

Bhagwati's idea: if a country's growth worsens its terms of trade enough, the loss can exceed the gain from growth. For India it would mean growth pushing up oil demand so much that the oil price rises against it. It needs very inelastic foreign supply and a heavy dependence on trade, so it is a warning, not India's actual situation.

B23. Why do freight costs matter for the terms of trade? (P2)

Imports are recorded CIF, including freight and insurance, while exports are recorded FOB, without them. When war-risk insurance or freight rates jump, as in the Red Sea in 2024 or Hormuz in 2026, India's measured import prices rise even if the oil price itself did not. That is lecture 14's transport-cost wedge, and we list it as a limitation.

B24. What is the Harberger–Laursen–Metzler effect? (P1)

A worse terms of trade lowers real income. If people smooth consumption, saving falls and the current account worsens. It is the link between our result and India's balance-of-payments stress in 1990–91, 2013 and 2022.

B25. Does Prebisch–Singer apply to India? (P1)

Prebisch–Singer says commodity exporters face a long-run decline in their terms of trade against manufactures. India is on the other side in oil: it imports the commodity. So falling commodity prices help it, which is what we see

in 2014–16, when the merchandise terms of trade rose 19%.

B26. Does the Marshall–Lerner condition matter here? (P1)

It says a depreciation improves the trade balance only if export and import demand are elastic enough. Oil demand is very inelastic, so a weaker rupee does little to cut the oil bill in the short run; it mainly raises the rupee cost. That is part of why oil shocks and rupee weakness come together, and why we model the price channel directly.

C. Literature

C1. Why so few references? (P2)

We kept it to the papers we use: four strands, about 16 core references. For a 15-minute talk, it is better to know each paper well than to list fifty. Every reference was checked against its DOI or official page.

C2. What is your anchor paper, and how do you differ from it? (P2)

Backus and Crucini (2000), "Oil prices and the terms of trade". They use a general-equilibrium model for industrial countries and find oil explains much of the variation in their terms of trade. We take the same question to India, which is an emerging importer and a refined-fuel exporter, and answer it empirically, with a theory benchmark and identified shocks.

C3. What did Kilian (2009) show? (P2)

That "not all oil price shocks are alike". He separates oil supply shocks, global demand shocks and oil-specific demand shocks, and shows they have different effects. Much of the 2000s price rise was demand-driven. It is the reason we ask H3.

C4. What do Baumeister and Hamilton (2019) do differently? (P3)

Kilian assumes oil supply cannot respond to price within a month. Baumeister and Hamilton replace that strict assumption with Bayesian priors on the supply and demand elasticities. They find supply disruptions matter more than Kilian's method implies. We use their shock series to split oil price changes.

C5. What is Känzig's oil supply news shock? (P4)

Känzig (2021, American Economic Review) measures how oil futures prices move in a narrow window around OPEC announcements. Because the window is so short, the move reflects only news about future supply. A negative news shock raises oil prices immediately. India cannot cause these shocks, so they are a clean test.

C6. What did Hamilton and Mork find about asymmetry? (P4)

Hamilton (1983) showed oil price rises came before most US post-war recessions. Mork (1989) found that rises hurt output but falls did not help as much. That is the basis of H2. For India's terms of trade we find no such asymmetry.

C7. What does Indian research say? (P2)

Bhanumurthy, Das and Bose (2012) trace oil shocks through import, price and fiscal channels. Ghosh (2011) finds higher oil prices depreciate the rupee. Tiwari and Olayeni (2013) show the oil–trade balance link changes across time horizons. Deheri and Sahu (2024) find oil-specific demand shocks worsen India's trade balances. None models the terms of trade directly.

C8. Deheri and Sahu say demand-type shocks matter for India. Does that agree with you? (P4)

Broadly yes. They look at the trade balance and find oil-specific demand shocks are the ones that matter. We look at the terms of trade and find demand-driven rises hurt at least as much as supply-driven ones. Both say the source of the shock does not make supply shocks the special danger for India.

C9. Schmitt-Grohé and Uribe find terms-of-trade shocks explain under 10% of output swings. So why care? (P2)

They ask how much output moves. We ask how much real purchasing power moves, which is the terms of trade itself. For India, terms-of-trade swings of 14–24% in a single oil episode are large real-income shocks, whatever they do to output in the short run.

D. Data

D1. Where does your data come from? (P2)

Only official agencies and the published data of peer-reviewed papers: the RBI Handbook of Statistics, DGCI&S, PPAC, the World Bank, UNCTAD, UN Comtrade, the US EIA and the BIS, plus the Baumeister–Hamilton and Känzig shock series. Every file is listed with its web address and a checksum in `data/SOURCES.md`.

D2. What is a unit value index, and how is it different from a price index? (P2)

A unit value is the value of a commodity divided by its quantity, for example rupees per tonne. DGCI&S combines them into Laspeyres indices with fixed base-year weights. A true price index would track the same product over time; a unit value also changes when the quality or mix within a commodity code changes. It is the best long official

measure for India, but it is not a pure price.

D3. Why two measures of the terms of trade? (P2)

The national-accounts measure (export deflator over import deflator, goods and services) goes back to 1970 on one method, so it covers all the big oil shocks. The merchandise measure (DGCI&S unit values) is the textbook goods definition but only links cleanly from FY1994. If both give the same answer, the result does not depend on one statistical source.

D4. Why do the two measures sometimes disagree, as in 2014–16? (P2)

The merchandise measure is goods only, where oil is about a quarter of imports. The goods-and-services measure adds services, about a third of India's exports, whose prices do not move with oil, so the oil effect is diluted. The two are also built differently: unit values versus national-accounts deflators.

D5. How did you handle the change in base years? (P2)

RBI publishes the merchandise series on 1978–79, 1999–2000 and 2012–13 bases. For each year we use the newest base available and link the older ones at the overlap year with a ratio. That keeps each base's growth rates and only rescales the level. It is the standard overlap-ratio method.

D6. Why not just use the old 1978–79 base series? It is longer. (P2)

Because it contradicts the newer base. Over FY1999 to FY2007, the 1978–79 base shows the terms of trade rising 21.7% while the 1999–2000 base shows it falling 21.0%. The old weights are from before India imported much oil or exported any refined fuel, so they misweight exactly the goods we study.

D7. You recompute the terms of trade instead of using RBI's published numbers. Why? (P2)

Because three of RBI's published values (FY2002–03, 2015–16 and 2018–19) do not equal RBI's own export index divided by its import index, beyond rounding. The definition in lecture 21 is the ratio of the two indices, so we compute it from them. Everywhere else, the recomputed and published values agree.

D8. How did you find the error in RBI Table 32? (P2)

The crude volumes for the 1990s were far too high: they matched the 2000s, when India imported several times more crude. On inspection, the eight rows for FY1990–91 to 1997–98 were an exact copy of FY2000–01 to 2007–08 in all four columns. We set them to missing, and the validation script now checks every table for repeated blocks of rows.

D9. How did you align fiscal years and calendar years? (P2)

Indian data are April to March. We average monthly world prices over April to March in code. For the MUV deflator, which is only calendar-year, we use 0.75 of year t plus 0.25 of year $t+1$, because nine months of the fiscal year fall in t . We also checked that WDI's "2024" for India is fiscal year 2024–25.

D10. Why do you use the real oil price, and why deflate by MUV? (P3)

The real price is what matters for the terms of trade: if all prices doubled, the terms of trade would not change. MUV is the World Bank's index of manufactured export prices, so oil divided by MUV is the price of oil in terms of manufactures. That is the relative price on the axes of the offer-curve diagram, and it is the World Bank's standard deflator for commodity prices.

D11. Why not use the Indian crude basket? (P2)

PPAC's usable monthly series only starts in 2020, and its website does not even have the FY2022–23 file. Our main price, the average of Brent, Dubai and WTI from the World Bank, goes back to 1960. The basket tracks Dubai and Brent closely, and our results are the same with Brent alone.

D12. What is the Indian crude basket? (P2)

PPAC's weighted average of the sour Gulf grades (Dubai and Oman) and sweet Brent-type crude, in the proportion India's refiners actually buy them. It is mostly sour crude, which is why it spiked more than Brent in March 2026: \$113 against \$104.

D13. Why do your monthly data only start in 2019? (P2)

DGCI&S's online archive for the 2012–13 base starts in FY2019–20, and the older 1999–2000 base page returns "No Data Found". We did not want to type numbers in from printed reports. So the monthly sample is 87 months, and the long annual series carry the history.

D14. Isn't 54 years too few for a time-series model? (P3)

It is small, which is why we chose ARDL (it behaves well in small samples), kept lags to at most 2, and used finite-sample critical values. The short-run result is significant at well below 1% in every specification, so sample size is not what drives it.

D15. Did anyone check the data by hand? (P2)

data/validation_log.md lists eleven values, at least one from every source and from old and recent years, with exactly where to find each in the original publication. One of us opens each source and ticks it off. The automatic checks pass with zero failures.

D16. How do you know World Bank and FRED oil prices are the same? (P2)

We compared them: correlation 0.9998 and a mean difference of 33 cents a barrel over 441 months. The PPAC basket lies between Dubai and Brent in 95% of normal months.

D17. Are your 2026 data final? (P2)

No. The latest months are provisional and may be revised. That is why the 2026 episode is shown as an illustration and the main estimates use the long annual series. We say on the limitations slide that the 2026 shock is still unfolding.

D18. What does a value like "FY2024" mean in your tables? (Any)

The fiscal year that starts in April 2024, so 2024-25. All our annual Indian data use that labelling.

E. Econometrics**E1. Why is everything in logs? (P3)**

So the coefficients are elasticities: the percentage change in the terms of trade for a 1% change in the oil price. That makes them comparable across periods and with the theory benchmark, which is also an elasticity. Logs also make the very large 1970s price jumps less dominant.

E2. What is a spurious regression? (P3)

If two series both wander or trend, a regression of one on the other can look highly significant even when they are unrelated, because both are just drifting. That is why we test the order of integration first and use a model built for non-stationary data.

E3. What does the ADF test do? (P3)

It tests whether a series has a unit root, meaning shocks to it are permanent. The null is a unit root; rejecting it means the series is stationary. "Augmented" means lagged differences are added to soak up serial correlation.

E4. Why also KPSS? (P3)

Its null is the opposite: the series is stationary. ADF has low power in short samples, so failing to reject a unit root is weak evidence on its own. Running both tells us where each series really stands. For us: levels are borderline, first differences are clearly stationary.

E5. What does Zivot–Andrews add? (P3)

A series with a one-time break, such as after 1991, can look like it has a unit root when it does not. Zivot–Andrews allows one break chosen by the data. It still does not reject a unit root for our levels (breaks found in 1985 for the terms of trade and 1999 for oil), so a single break is not what drives our results.

E6. Why ARDL rather than OLS in levels? (P3)

OLS in levels risks a spurious regression with non-stationary data. OLS in differences throws away the long-run information. ARDL in error-correction form keeps both: the short-run effect from the differences and the long-run relationship from the lagged levels.

E7. Why ARDL and not Johansen? (P3)

Johansen needs all variables to be $I(1)$, and ours are borderline between $I(0)$ and $I(1)$. The ARDL bounds test is valid either way. It also works better in small samples like our 54 years, and we have a clear dependent variable, the terms of trade, so a single equation is natural.

E8. How did you choose the lags? (P3)

By the Akaike information criterion, searching up to two lags of each variable. It picked ARDL(1,1,1): one lag of the terms of trade and one of each regressor. More than two lags with 54 observations would use up too many degrees of freedom.

E9. What does $\alpha = -0.31$ mean? (P3)

It is the speed of adjustment. If the terms of trade is away from its long-run relationship with oil, 31% of the gap closes in the next year. That is a half-life of about two years. It has the right sign (negative) and is between -1 and 0 , which a valid error-correction model needs.

E10. Explain the bounds test and your result. (P3)

It is an F-test that all the lagged levels are jointly zero, which means there is no long-run relationship. The critical values come as a lower bound (all regressors $I(0)$) and an upper bound (all $I(1)$). Our F of 4.74 is above the 10% upper bound of 4.32 but between the 5% bounds of 4.02 and 5.13. So there is a long-run relationship at 10%, and the 5% test is inconclusive.

E11. Why finite-sample critical values? (P3)

The original Pesaran–Shin–Smith critical values are for large samples. With 54 observations they are too lenient: you would find cointegration too often. Kripfganz and Schneider (2020) give critical values and p-values for the

actual sample size, so ours are the stricter, more honest version.

E12. Which case of the bounds test did you use? (P3)

Case 3: an unrestricted intercept and no trend. The terms of trade has no significant trend over 1970–2025 (110 in FY1970, 98 in FY2025), so an intercept is enough.

E13. The long-run elasticity is not significant. Doesn't that undermine H1? (P3)

No. H1 is about whether oil rises worsen the terms of trade, and the short-run elasticity of -0.24 is significant at well below 1% in every specification. What the long run tells us is that the effect is temporary: the economy largely adjusts. We say that openly instead of overselling a long-run result.

E14. How do you get a standard error for the long-run elasticity? (P3)

It is a ratio of two estimates, $-\beta_1/\alpha$, so we use the delta method: a first-order Taylor approximation that combines the variances and the covariance of β_1 and α . It gives 0.059.

E15. You found heteroskedasticity. What does it change? (P3)

Heteroskedasticity does not bias the coefficients, only the usual standard errors. So we also report HC1 robust standard errors. The short-run standard error goes from 0.049 to 0.040, so the result gets, if anything, stronger.

E16. What do CUSUM and CUSUMSQ show? (P3)

They track the cumulative sum of recursive residuals (and of their squares). If the coefficients changed during the sample, the line would cross the 5% bands. Ours stay inside, so the relationship is stable over 1970–2025.

E17. What does the RESET test check? (P3)

Whether powers of the fitted values add explanatory power, which would mean the functional form is wrong (for example, a missing non-linearity). $p = 0.84$, so there is no sign of that.

E18. What is the R^2 of your ARDL, and is it low? (P3)

0.51 (adjusted 0.45) for an equation in first differences, explaining year-to-year changes in the terms of trade. For annual changes that is respectable, and the question is not whether we explain everything but whether the oil effect is significant and stable.

E19. Isn't the oil price endogenous? (P3)

Partly, because India is a big buyer. In annual data India's own demand moves world prices only slowly, and the oil price is mostly driven by world supply and demand. We handle it directly in H3 with Baumeister–Hamilton and Känzig shocks, which India cannot cause, and those give the same picture.

E20. Which controls did you use, and why not more? (P3)

The real non-energy commodity price, which separates oil from a general commodity boom. A robustness check adds the gold price. With 54 years we cannot add many controls, and some, like the exchange rate, respond to oil themselves. Controlling for them would remove part of the effect we want to measure.

E21. Oil and other commodity prices move together. Is that multicollinearity? (P3)

They are correlated, about 0.47 in annual changes, but far from perfectly. The test is whether the oil coefficient is stable when we drop the control: it is about -0.20 without it and -0.24 with it, both highly significant.

E22. Explain NARDL. (P4)

It splits the oil price into two running totals, one of all the increases and one of all the decreases, and puts both in the ARDL. Then a rise and a fall can have different coefficients, and a Wald test checks whether they are equal, separately for the short run and the long run.

E23. How do you read the signs in NARDL? (P4)

The "falls" total only ever goes down. A negative coefficient on it means a fall in oil prices raises the terms of trade. So both coefficients negative means "rises hurt, falls help", and asymmetry is about whether they have the same size.

E24. Why do you call the long-run asymmetry "not robust"? (P4)

In the baseline it is significant at 5% ($p = 0.035$), but it disappears with small changes: $p = 0.05$ with one more lag, 0.08 from 1975, 0.28 from 1980, and it reverses in the merchandise data. A result that depends on including the 1970s is not something we would base policy on.

E25. What is a local projection? (P3)

Instead of estimating one dynamic model and simulating it forward, like a VAR, you run a separate regression for each horizon: the change in the terms of trade from month $t-1$ to month $t+h$ on the oil price change in month t . The coefficient at each horizon is the cumulative response. It is Jordà's 2005 method.

E26. Why local projections and not a VAR? (P3)

With about 85 months, a VAR's impulse responses depend on getting the whole lag structure right, and an error at one lag affects every horizon. Local projections estimate each horizon directly and are more robust to that. The cost is wider confidence bands.

E27. Why HAC standard errors with $h + 1$ lags? (P_3)

At horizon h , the left-hand side overlaps across consecutive months, so the errors are autocorrelated by construction, up to h lags. Newey–West (HAC) standard errors correct for that.

E28. Explain the H3 method. (P_4)

Two steps. First, regress each month's oil price change since 1975 on Baumeister and Hamilton's four structural shocks and their lags. The part explained by supply shocks is the supply-driven change; the part explained by the activity, consumption and inventory shocks is the demand-driven change. Second, regress India's terms-of-trade changes on the two parts and test whether their coefficients are equal.

E29. Isn't there a generated-regressor problem? (P_4)

Yes, in principle: the two parts are estimated, so second-step standard errors are a bit too small. But the first step is very precise (612 months, R^2 of 0.80), and our conclusion is that the two effects are *not* different. Correcting the standard errors would make them larger, which only strengthens that conclusion.

E30. What does $R^2 = 0.80$ in the first step mean? (P_4)

The four structural shocks explain 80% of the month-to-month variation in the oil price since 1975. So the split into supply and demand parts covers most of what moves oil.

E31. Why are Känzig's shocks exogenous to India? (P_4)

They are measured from oil futures price moves in a very short window around OPEC announcements. Nothing India does in that window causes the move. So when India's terms of trade falls after one of these shocks, the direction of cause is clear.

E32. How do you deal with structural breaks, like the 1991 reforms? (P_3)

Three ways. Zivot–Andrews allows a break in the unit-root test. CUSUM and CUSUMSQ show the coefficients are stable. And the post-1980 subsample and the rolling regressions show how the effect changed. The rolling estimate changes gradually, as refining capacity grew, not in a sudden jump.

E33. The rolling windows overlap. Can you test the decline? (P_4)

Overlapping windows are not independent, so we treat the rolling chart as descriptive. The test is that the decline matches the theory benchmark built from trade shares, which is completely separate data, and that the first and last windows (1970–89 and 2005–24) do not overlap at all: -0.30 (s.e. 0.09) against -0.11 (s.e. 0.06).

E34. Why do the monthly and annual elasticities differ (-0.39 vs -0.24)? (P_3)

They measure different things. The monthly one is merchandise only, where oil is a bigger share, and the plain merchandise elasticity in annual data is -0.30 , close to it. The annual goods-and-services measure is diluted by services. Annual averaging also smooths short spikes.

F. Results**F1. Interpret -0.24 in plain words. (P_3)**

If the real oil price rises 10% this year, India's terms of trade fall about 2.4% this year: each unit of India's exports buys 2.4% less imports than before.

F2. Why is the short-run effect bigger than the long-run effect? (P_3)

In the short run, contracts and demand are fixed, so the higher oil price passes straight into import prices. Over time, refined-fuel export prices catch up with crude, the export mix and the rupee adjust, and oil spikes tend to reverse. That is lecture 13's time-horizon logic.

F3. Why does the effect take two to three months? (P_3)

Crude cargoes are priced on the previous month's benchmark and recorded at customs when they arrive, often weeks after loading. So a price rise in March shows up in India's import unit values in April and May.

F4. Why did India's terms of trade not fall immediately in March 2026? (P_4)

Because India's refined-fuel exports rose in price too. From February to April 2026, export unit values rose 18.1% and import unit values 15.7%. By June, import prices had kept rising and export prices had fallen back, so the terms of trade was 13.1% lower. The refining hedge delayed the hit but did not stop it.

F5. What did 2022 look like? (P_4)

From February to June 2022, Brent rose 25% and the merchandise terms of trade fell 16.1%, to its lowest point in our monthly data (89.4 in June 2022). In fiscal-year averages, FY2022 was 14.4% below FY2020 for goods and services and 23.7% below for merchandise.

F6. Why does 2008 show an improvement? (P_4)

The 2008 spike lasted only months: oil peaked in July and collapsed by December. A fiscal-year average barely captures it, and the following crash falls in the same fiscal year. Monthly data would show it, but DGCI&S monthly

data for 2008 are not online.

F7. How sure are you that exposure has halved? (P₄)

Two independent pieces of evidence agree. The rolling regression goes from -0.30 to -0.11 , and the trade-share benchmark, built from completely different data, goes from -0.25 to -0.10 . And the mechanism is visible in the RCA for refined fuels, which went from 0.06 to 4.37 .

F8. H2 was not supported. Does that mean oil shocks are never asymmetric? (P₄)

No. Mork's asymmetry is about output, where firms can delay investment when oil prices are volatile. The terms of trade is a price ratio that passes oil prices through almost mechanically in both directions. So it is quite consistent for output to be asymmetric and the terms of trade to be symmetric.

F9. H3 was not supported. Were you surprised? (P₄)

At first, yes, because the offer-curve story suggested a boom would help India's exports. But a world boom raises the prices of all commodities, and India imports many of them: gold, coal, fertiliser, edible oils, metals. India's exports are mostly manufactures and services, whose prices rise less. So demand-driven rises hurt at least as much.

F10. Two of your three hypotheses were rejected. Did the project fail? (P₁)

No. A hypothesis test is supposed to be able to fail. We learned that what matters for India is how much oil rises, not why, and that rises and falls matter equally, which changes the policy advice: India needs symmetric buffers that work whatever the cause of the shock.

F11. What surprised you most? (Any)

That India's exposure has halved, and that the simple trade-share formula from the definition of the terms of trade tracks it so closely. We expected refining to matter, but not that a one-line formula would get the size right.

F12. How reliable is the monthly -0.39 with only 85 months? (P₃)

The bands are wide, but the effect is significant at every horizon from one to six months. For example, at two months it is -0.39 with a standard error of 0.13 , and at three months -0.37 with 0.08 . Month 0 is close to zero, which is also what we expected.

F13. What does the benchmark coefficient of 1.70 tell us? (P₂)

That the trade-share arithmetic gets the direction and order of magnitude right, and we cannot reject that it is exactly right ($p = 0.07$). The point estimate above 1 says the real effect is a bit bigger than the direct oil channel, because of the indirect cost of oil in fertiliser, petrochemicals and freight.

F14. If oil prices fell 30% tomorrow, what would happen to India's terms of trade? (P₃)

Since effects are symmetric, roughly $0.24 \times 30 \approx 7\%$ better in the goods-and-services measure within a year, and more in the merchandise measure, mostly within two to three months. Then part of it would fade as the economy adjusts.

F15. What would your model say about FY2026? (P₄)

It depends on how long prices stay high. The monthly data already show the merchandise terms of trade 13% lower in June than in February. If the fiscal-year average oil price ends well above FY2025's, the annual model points to a fall of roughly a quarter of that percentage rise, with recovery in the following years as prices normalise.

G. Policy and current affairs

G1. What should India do? (P₄)

Four things. Build buffers that work both ways, such as strategic reserves and a fuel-tax buffer, because rises and falls matter equally. Keep growing the refining hedge. Diversify crude suppliers to weaken any one seller's market power. And cut overall commodity dependence through the energy transition, because demand-driven booms hurt too.

G2. What is a fuel-tax buffer? (P₄)

Cut fuel duties when oil spikes and rebuild them when oil is cheap, so consumers face smoother prices and the government's revenue absorbs the swing. India did a version of this: excise duties were raised in 2020 when oil crashed and cut in November 2021 and May 2022 when it spiked.

G3. Does India have strategic oil reserves? (P₄)

Yes: underground caverns at Visakhapatnam, Mangaluru and Padur, about 5.3 million tonnes, which is roughly 9–10 days of consumption. Together with the oil companies' own stocks, the cover is longer. Our results say reserves are worth more than they look, because they help during the two-to-three-month window when the terms-of-trade hit lands.

G4. Did buying Russian crude after 2022 help? (P₄)

It is the pro-competitive effect from lecture 16: more suppliers and a discount. Because DGCI&S unit values record the prices India actually paid, any discount is already in our import price index. We did not estimate its size separately, so we would not put a number on it.

G5. Should India restrict refined-fuel exports during a spike to protect domestic supply? (P_4)

That would weaken the hedge. Refined exports are what keep India's export prices rising with oil, which is why exposure has halved. Restricting them would raise $s_m - s_x$ again. There is a domestic-supply trade-off, but in terms of the terms of trade, export restrictions make things worse.

G6. What about windfall taxes on fuel exports, like in 2022? (P_4)

India introduced special duties on fuel exports and domestic crude in July 2022. They raise revenue, but taxing fuel exports reduces what India earns from the refining hedge during the very period it matters most. Our results would argue for keeping such taxes temporary.

G7. How does ethanol blending fit in? (P_4)

Blending ethanol into petrol replaces some imported crude with a domestic product, which lowers s_m . And since we find demand-driven commodity booms hurt too, reducing commodity-import dependence in general, through renewables and electric transport, is the long-run answer.

G8. Would paying for oil in rupees help? (P_4)

It reduces exposure to the dollar, but not to the oil price. The terms of trade is a ratio of prices; whether India pays in rupees or dollars, a 46% jump in the oil price still means each unit of exports buys less oil.

G9. How did the Red Sea attacks in 2024 matter? (P_2)

Ships rerouted around Africa, and freight and insurance costs rose. Since India's imports are valued including freight (CIF) and exports without it (FOB), that raises measured import prices and worsens the measured terms of trade, even without an oil price change. That is lecture 14's transport-cost wedge.

G10. What does your work say about the rupee in 2026? (P_4)

We do not model the exchange rate directly. The record low of 94.65 per dollar at end-March 2026 is consistent with the literature (Ghosh 2011): oil shocks raise India's import bill, which weakens the rupee. A weaker rupee does not change the terms of trade much, because both export and import prices are mostly set in dollars.

G11. If you advised the RBI, what would you say? (P_4)

Expect most of the terms-of-trade hit within two to three months of an oil spike, about 0.25% per 1% rise in the short run, and expect the same size of gain when oil falls. Use the gains in good times to build reserves.

G12. Does the energy transition make this question irrelevant? (P_4)

Not soon. Oil is still about a quarter of India's imports, and India's oil demand is growing. Over decades, lower oil use would shrink s_m and the exposure, which is the same direction refining has already moved it.

H. Limitations and extensions

H1. What is your biggest limitation? (Any)

Unit value indices are not true price indices: a change in the mix or quality of goods within a commodity code moves them too. It affects every study of India's terms of trade, and both of our measures (unit values and national-accounts deflators) give the same answer.

H2. What else would you list? (Any)

The monthly sample is short (87 months). The goods-and-services measure dilutes oil with services. CIF/FOB valuation lets freight shocks move the measured terms of trade. And 2026 data are provisional.

H3. Could gold be driving your results? (P_3)

Gold is a big Indian import and its price moves with commodity booms. So one robustness check adds the real gold price. The short-run oil elasticity is -0.22 with gold included, still highly significant.

H4. What would you do with more time? (Any)

Digitise older DGCI&S monthly reports to extend the monthly series back before 2019. Use commodity-level unit values to split the effect into crude, products and other commodities. Use Känzig's shock as an instrument for the oil price in the local projections (LP-IV). And link the terms-of-trade effect to the current account, the HLM channel.

H5. Do your results apply to other countries? (Any)

The method does, the numbers do not. The benchmark ($s_x - s_m$) tells you what to expect for any country: an oil importer with no refining exports, like Bangladesh, should be more exposed; one that also exports products, like South Korea, should look more like India today.

H6. Isn't the refining "hedge" just mechanical, since fuel prices follow crude? (P_2)

Yes, and that is exactly why it works. Because refined-fuel export prices move with crude, a bigger fuel export share

means more of each oil shock comes back to India as higher export prices. The value added on top (the 1.2 to 1.4 unit-value ratio) is what India earns from refining.

H7. You did not include the exchange rate. Isn't that a missing variable? (P3)

The terms of trade is a ratio of two prices that are mostly set in dollars, so the exchange rate largely cancels out. And the exchange rate reacts to oil shocks itself, so controlling for it would absorb part of the oil effect we want to measure.

H8. Why not use a VAR for the annual data? (P3)

With 54 years, a VAR with three or four variables and two lags would have too many parameters for the data. ARDL is much more economical and answers our question directly.

I. Curveballs

I1. Can you draw the offer-curve diagram for us? (P1)

Axes: India's exports X across, oil Y up. Draw India's curve bending up towards the Y axis and the rest of the world's curve bending towards the X axis; mark where they cross and draw the ray from the origin. For a supply cut, rotate the rest-of-world curve towards the X axis; the new crossing has a flatter ray, which is India's worse terms of trade.

I2. What is the difference between the terms of trade and the real exchange rate? (P1)

The terms of trade compares India's export prices with its import prices. The real exchange rate compares India's overall price level, or its tradable goods, with other countries' prices, adjusted for the nominal exchange rate. An oil shock worsens the terms of trade directly; the real exchange rate moves through the rupee and inflation.

I3. Your slides and tables show different standard errors for the same number. Why? (P3)

The table shows both the usual and the heteroskedasticity-robust (HC1) standard errors. For the short-run elasticity they are 0.049 and 0.040. The conclusions are the same with either.

I4. What software did you use? (P3)

Python: pandas for data, statsmodels for ARDL, OLS, HAC and local projections, the arch package for unit-root tests, and matplotlib for charts. The slides are LaTeX Beamer. Everything is in the repository and runs from the raw files.

I5. Can someone reproduce your results? (P3)

Yes. The raw files are in the repository exactly as downloaded, with checksums. One script builds the dataset, another validates it, and the notebooks produce every table and chart. The notebooks are saved with their outputs, so they can be read without running anything.

I6. Which lecture was most useful? (P1)

Lecture 22 on offer curves, because it gave us the hypotheses. The benchmark comes from lecture 21's definition of the terms of trade, and lectures 8, 17, 19 and 20 explain why India's exposure fell.

I7. Where do the Leontief paradox or factor price equalisation fit in? (Any)

They don't, directly, and we did not force them in. Those lectures are about the pattern of trade and factor returns; our question is about the price of trade. The parts of H-O we do use are endowments (India is oil-scarce) and the time horizons from lecture 13.

I8. Your hypotheses came from theory, but two failed. Is the theory wrong? (P1)

The theory was ambiguous, not wrong. For H3, offer curves say a world boom shifts both curves, so the sign was always empirical. For H2, the asymmetry idea comes from studies of output, not of prices. What theory did predict clearly, H1 and its size through the benchmark, the data confirm.

J. When you don't know the answer

1. **Pause and restate the question** in your own words. It buys time and makes sure you answer what was asked.
2. **Answer the part you know** and connect it to our results: "I'm not sure about X, but in our data, Y."
3. **Be honest:** "We didn't test that directly. Our closest evidence is ..." Examiners respect that far more than a bluff.
4. **Hand over once, cleanly:** "P3 ran the econometrics, so I'll let them add to this." Then stop talking. Don't hand over twice.
5. **Never invent a number.** If you don't remember it, give the direction and say where it is: "It's in table 6 of the backup slides."

O6 · Formula sheet

Everything on one page. $y = \ln \text{ToT}$, $x = \ln \text{real oil price}$, $z = \ln \text{real non-energy commodity price}$.

Trade measures

Measure	Formula	Lecture
Net barter terms of trade	$\text{NTT} = (\text{UVI}_X / \text{UVI}_M) \times 100$	L21
Income terms of trade	$\text{ITT} = \text{NTT} \times \text{QI}_X / 100$	–
Gross barter terms of trade	$\text{GBT} = (\text{QI}_M / \text{QI}_X) \times 100$	–
Revealed comparative advantage (Balassa)	$\text{RCA}_{ij} = (X_{ij} / X_i) / (X_{wj} / X_w)$; $\text{RCA} > 1 = \text{advantage}$	Lo8
Grubel-Lloyd index	$\text{GL} = 1 - X - M / (X + M)$; 0 = one-way, 1 = fully two-way	L19
Horizontal vs vertical IIT	Horizontal if $1/1.15 \leq \text{UV}_X / \text{UV}_M \leq 1.15$; otherwise vertical	L20
Monopoly markup (Lerner)	$\text{MR} = p(1 - 1/e) = \text{MC}$, so $(p - \text{MC})/p = 1/e$	L16
Average cost with scale economies	$\text{AC} = F/Q + c$	L16
Real oil price	$\text{P}_{\text{oil}} / \text{MUV} \times 100$	–
Fiscal-year MUV	$0.75 \times \text{MUV}_{\text{CY } t} + 0.25 \times \text{MUV}_{\text{CY } t+1}$	–

The theory benchmark

$$d \ln \text{NTT} \approx (s_x - s_m) \cdot d \ln \text{P}_{\text{oil}}$$

s_x = oil exports / total exports; s_m = oil imports / total imports. Benchmark -0.25 (FY1999) $\rightarrow -0.10$ (FY2025).

ARDL(1,1,1) in error-correction form (H1)

$$\Delta y_t = c + \alpha y_{t-1} + \beta_1 x_{t-1} + \beta_2 z_{t-1} + \gamma_1 \Delta x_t + \gamma_2 \Delta z_t + \varepsilon_t$$

Quantity	Formula	Our value
Short-run elasticity	γ_1	–0.241
Long-run elasticity	$-\beta_1 / \alpha$	–0.092
Speed of adjustment	α (must be between -1 and 0)	–0.308
Half-life of a deviation	$\ln 0.5 / \ln(1 + \alpha)$	≈ 1.9 years
Bounds test	F-test of $\alpha = \beta_1 = \beta_2 = 0$ (case 3)	$F = 4.74$
Delta-method variance of $-\beta_1 / \alpha$	$g' V g$, with $g = (-1/\alpha, \beta_1/\alpha^2)$	s.e. 0.059

Finite-sample bounds ($N = 54$, $k = 2$): 10% 3.32 / 4.32; 5% 4.02 / 5.13; 1% 5.69 / 7.03.

NARDL (H2)

$$x^+ t = \sum_{j \leq t} \max(\Delta x_j, 0), \quad x^- t = \sum_{j \leq t} \min(\Delta x_j, 0)$$

$$\Delta y_t = c + \alpha y_{t-1} + \theta^+ x^+_{t-1} + \theta^- x^-_{t-1} + \pi^+ \Delta x^+_t + \pi^- \Delta x^-_t + \text{controls} + \varepsilon_t$$

Long run $L^\pm = -\theta^\pm / \alpha$. Wald tests: $L^+ = L^-$ (long run), $\pi^+ = \pi^-$ (short run).

Benchmark test

$$\Delta \ln \text{NTT}_t = a + b [(s_x - s_m)_{t-1} \Delta \ln \text{P}_{\text{oil},t}] + c \Delta \ln \text{P}_{\text{nonoil},t} + e_t, \quad H_0: b = 1$$

Result: $b = 1.70$ (s.e. 0.38), $p(b = 1) = 0.07$, $N = 30$.

Local projections (Jordà 2005)

$$y_{t+h} - y_{t-1} = a_h + \beta_h \Delta \ln \text{Brent}_t + \varphi_h \Delta \ln \text{Brent}_{t-1} + \psi_h \Delta y_{t-1} + u_{t+h}, h = 0 \dots 6$$

HAC (Newey–West) standard errors with $h + 1$ lags. $\beta_2 = -0.39$.

Supply vs demand split (H3)

Step 1: $\Delta \ln P_{\text{oil},t} = c + \sum_k \sum_{j=0}^2 b_{kj} \varepsilon_{k,t-j} + u_t$, with $k \in \{\text{supply, activity, consumption demand, inventory demand}\}$. $R^2 = 0.80$, 612 months.

- Supply part_t = $\sum_j b_{\text{supply},j} \varepsilon_{\text{supply},t-j}$
- Demand part_t = the same sum over the other three shocks

Step 2: $\Delta \text{ToT}_t = a + b_S \cdot \text{supply part}_t + b_D \cdot \text{demand part}_t + e_t$, $H_0: b_S = b_D$.

Tests and what their null hypothesis is

Test	Null	Our p
ADF, Phillips–Perron	Unit root	levels ≈ 0.05 , differences 0.000
KPSS	Stationary	levels 0.01–0.10
Zivot–Andrews	Unit root with one break	0.39 (ToT), 0.53 (oil)
Breusch–Godfrey	No serial correlation	0.90
Breusch–Pagan	Homoskedastic errors	0.019
Jarque–Bera	Normal errors	0.26
Ramsey RESET	Correct functional form	0.84
CUSUM / CUSUMSQ	Stable coefficients	inside 5% bands

Standard errors

Type	Robust to	Used in
HC1 (White)	Heteroskedasticity; small-sample factor $n/(n - k)$	ARDL robust table, rolling regressions
HAC (Newey–West)	Heteroskedasticity and autocorrelation	Benchmark test, local projections, H3